But this is only one example. The more you think about it, the more you realize that life is riddled with Iterated Prisoner's Dilemma games, not just human life but animal and plant life too. Plant life? Yes, why not? Remember that we are not talking about conscious strategies (though at times we might be), but about strategies in the 'Maynard Smithian' sense, strategies of the kind that genes might preprogram. Later we shall meet plants, various animals, and even bacteria, all playing the game of Iterated Prisoner's Dilemma. Meanwhile, let's explore more fully what is so important about iteration.

Unlike the simple game, which is rather predictable in that DEFECT is the only rational strategy, the iterated version offers plenty of strategic scope. In the simple game there are only two possible strategies, COOPERATE and DEFECT. Iteration, however, allows lots of conceivable strategies, and it is by no means obvious which one is best. The following, for instance, is just one among thousands: 'cooperate most of the time, but on a random 10 per cent of rounds throw in a defect'. Or strategies might be conditional upon the past history of the game. My 'grudger' is an example of this; it has a good memory for faces, and although fundamentally cooperative it defects if the other player has ever defected before. Other strategies might be more forgiving and have shorter memories.

Clearly the strategies available in the iterated game are limited only by our ingenuity. Can we work out which is best? This was the task that Axelrod set himself. He had the entertaining idea of running a competition, and he advertised for experts in games theory to submit strategies. Strategies, in this sense, are preprogrammed rules for action, so it was appropriate for contestants to send in their entries in computer language. Fourteen strategies were submitted. For good measure Axelrod added a fifteenth, called Random, which simply played COOPERATE and DEFECT randomly, and served as a kind of baseline 'non-strategy': if a strategy can't do better than Random, it must be pretty bad.

Axelrod translated all 15 strategies into one common programming language, and set them against one another in one big computer. Each strategy was paired off in turn with every other one (including a copy of itself) to play Iterated Prisoner's Dilemma. Since there were 15 strategies, there were 15 × 15, or 225 separate games going on in the computer. When each pairing had gone through 200 moves of the game, the winnings were totalled up and the winner declared.